



Claude & Codex Skills For Academics

By Ilya Shabanov • [EffortlessAcademic.com](https://effortlessacademic.com)

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<https://effortlessacademic.com/ai-skills-for-academics/>

Tools enabled humans to funnel their (limited) strength to a narrow purpose. A knife cuts, an arrow flies, a pot stores.

Skills are the equivalent of tools in the AI age - They funnel AI's capabilities to a narrow purpose, so that it can solve specific problems far better than without it.

Technically speaking, a skill is a text file with instructions and resource files. This makes it easy to distribute and create skills.

But, because everyone can create skills, there are also a lot of less useful skills out there.

This guide exists to give you tried-and-tested skills.

AI and digital tools enabled me to write 4 papers and build an educational business on the side during my PhD, and still finish ahead of schedule.

I think of AI as musicians, and you are a conductor. If your musicians are skilled high schoolers, the orchestra will be okay, but not overwhelming.

However, by adding various skills, you are working with professional musicians. The challenge now shifts to the music itself and your interpretation as the conductor.

**Hi there, my name is Ilya Shabanov,
and I am The Effortless Academic. I will
Introduce myself properly at the end of
this document!**





Part 1: Tutorial on Installing & Downloading Skills

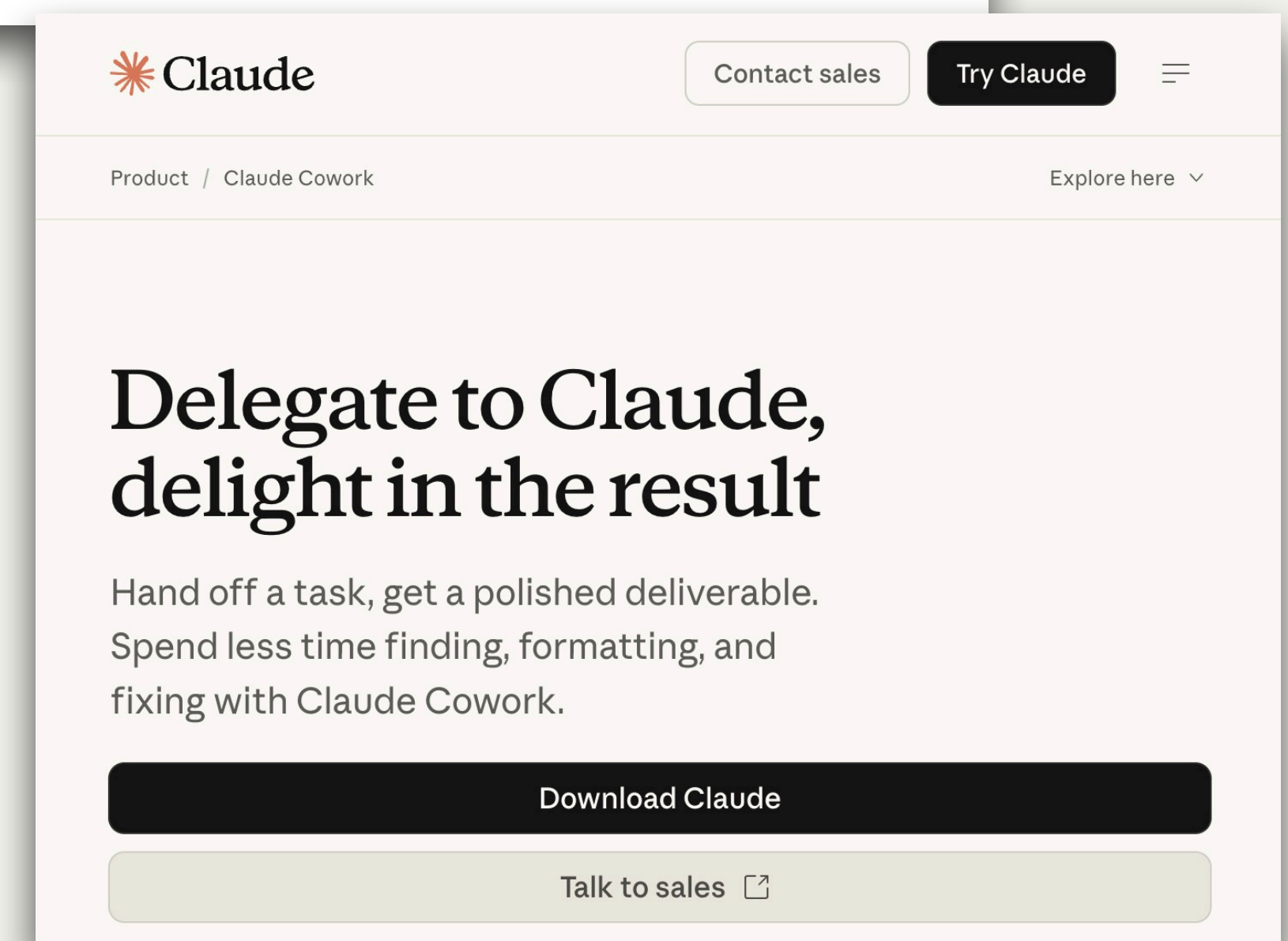
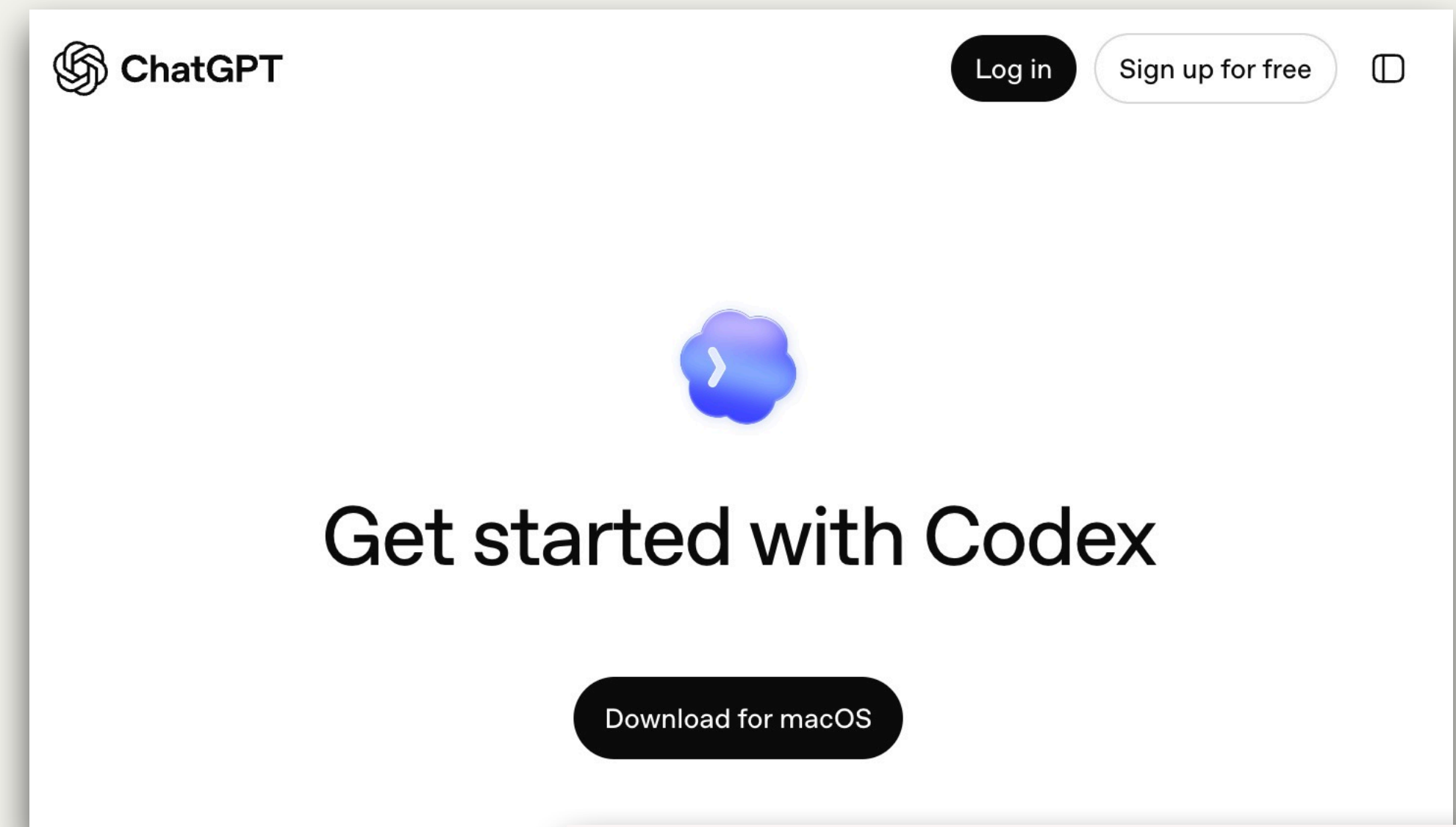
Pre Requisites: Claude Code or OpenAI Codex

To use these skills, you need to install either the Claude Desktop app or ChatGPT Codex.

Both bring AI to your desktop and give it access to your local files. This makes working with AI faster and more convenient, as you can merge it with your notes, papers, and so on without needing to upload them.

Skills presented here will work on both platforms.

Go ahead and download these to continue.

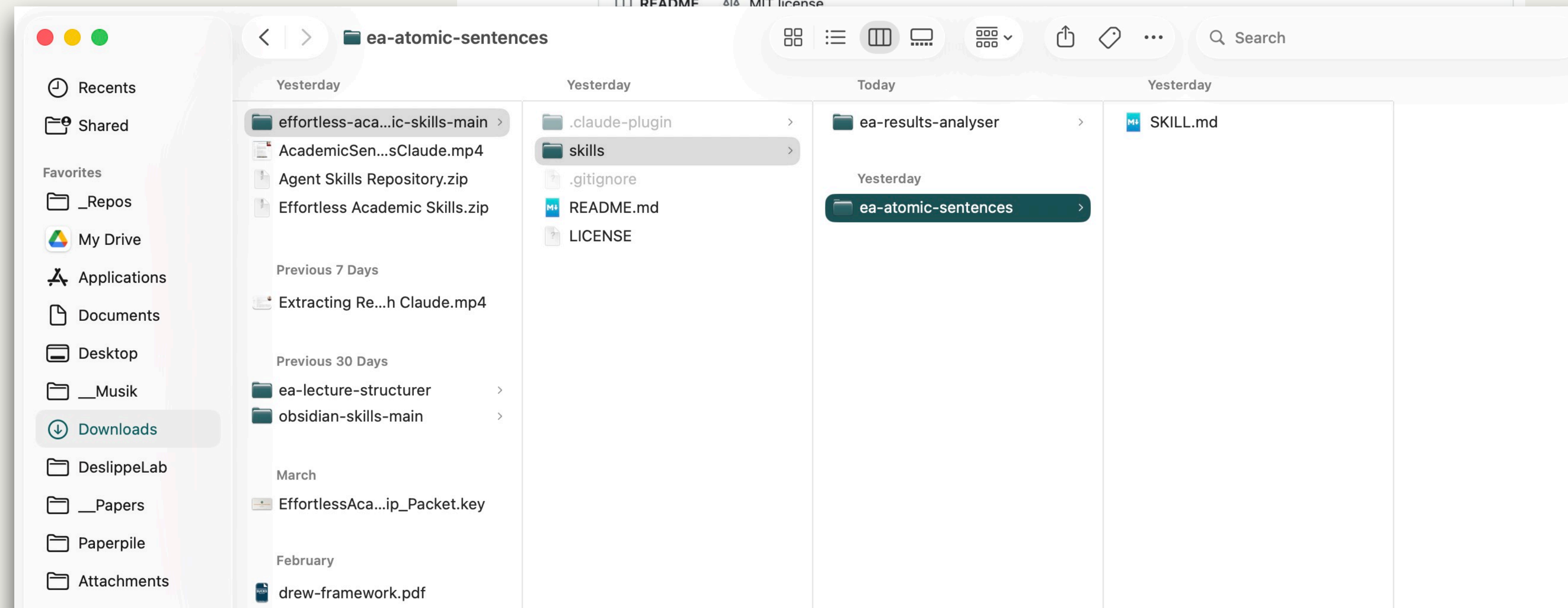
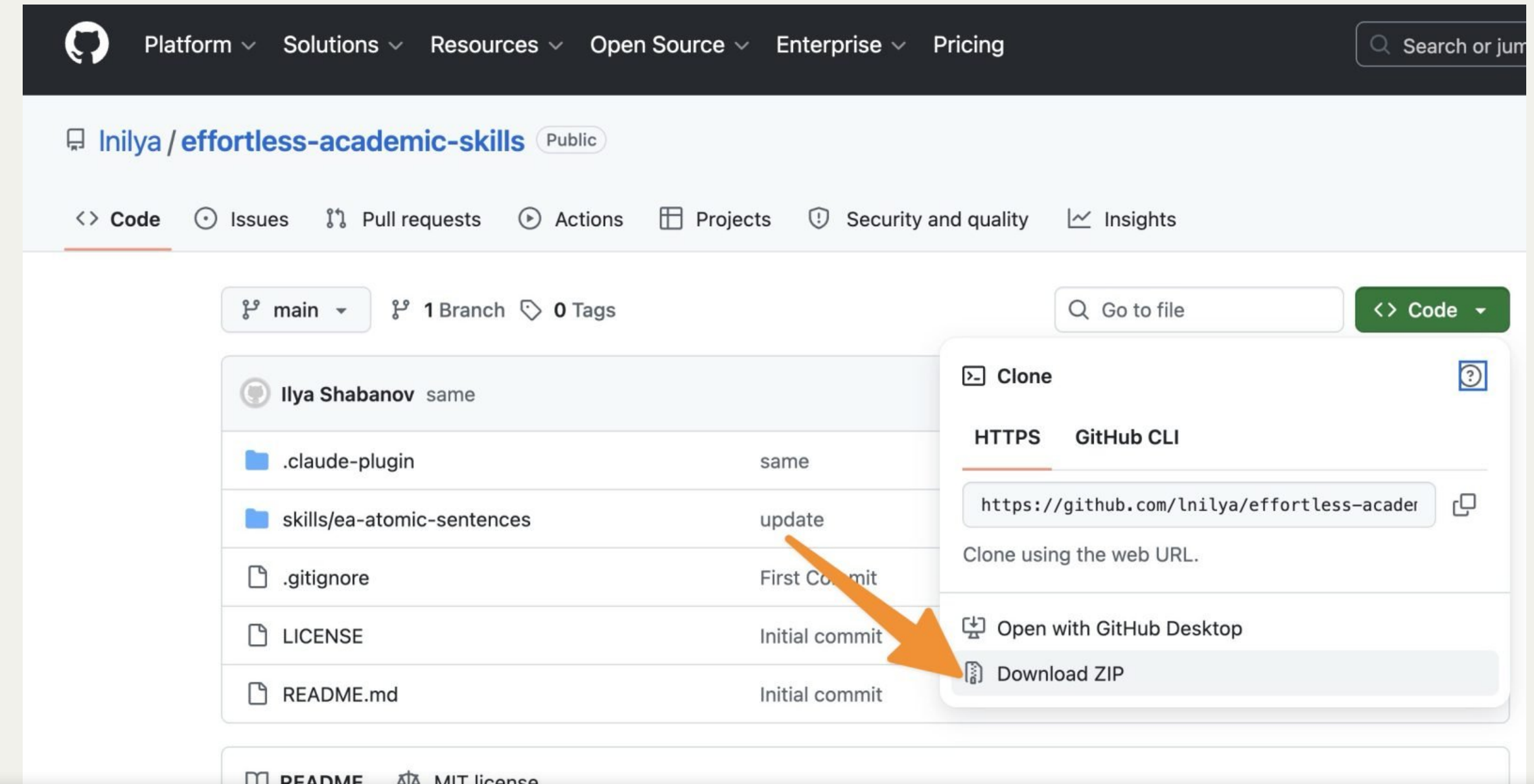


In this tutorial, you will get download links. These links typically go to GitHub, a service for versioning and storing files and code.

You don't need technical skills to use it.

Simply download the entire repository and unpack the downloaded zip file. Then navigate to the skills folder.

A skill is a folder that contains a skill.md file and possibly other elements.



Installing Skills - Open AI Codex

Open AI Codex stores skills inside a (hidden) folder:

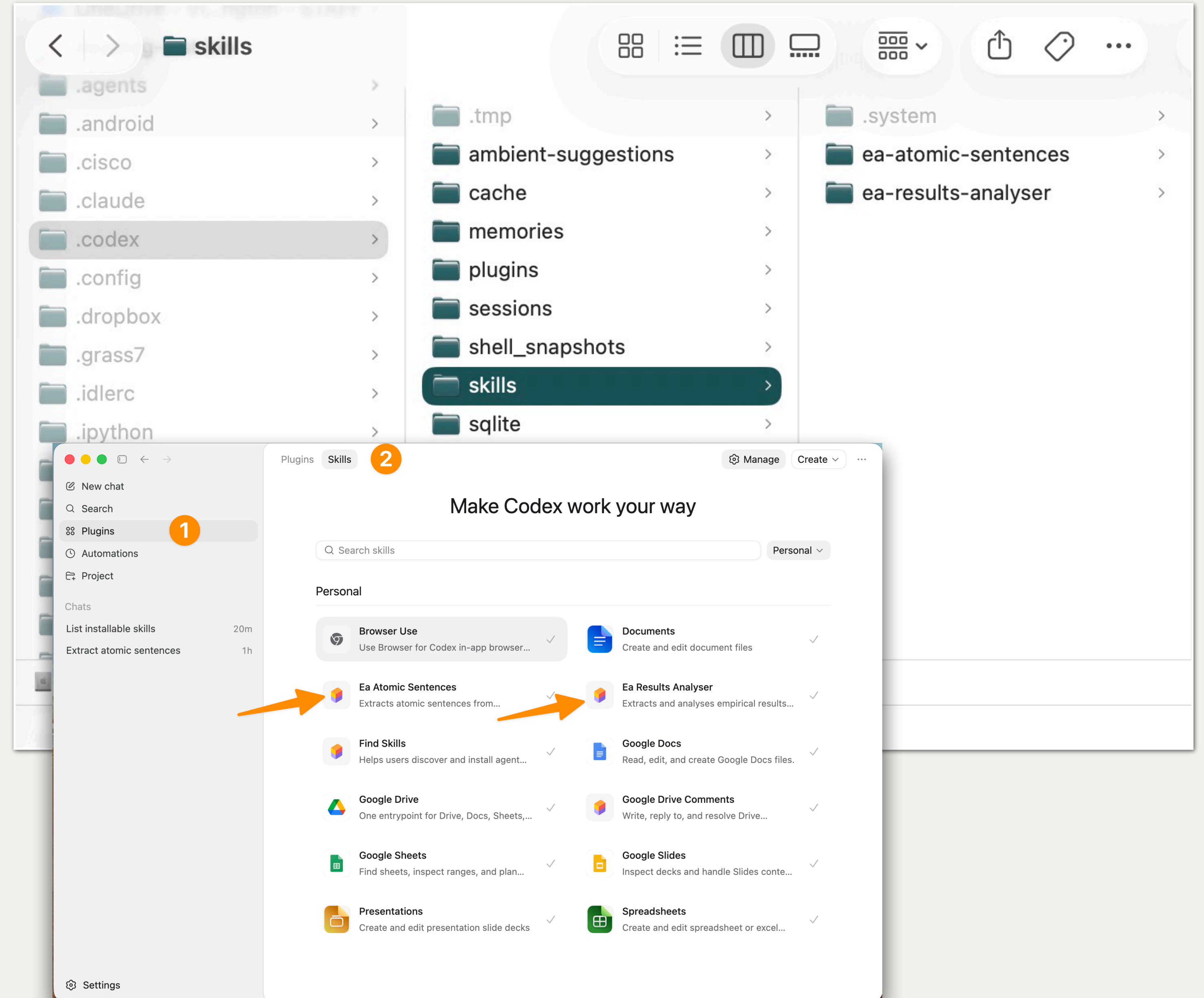
Mac OS: “Users/<username>/.codex/skills/”

Windows: “%USERPROFILE%\codex\skills\”

All you need to do is copy the skill folder here.

Restart Codex

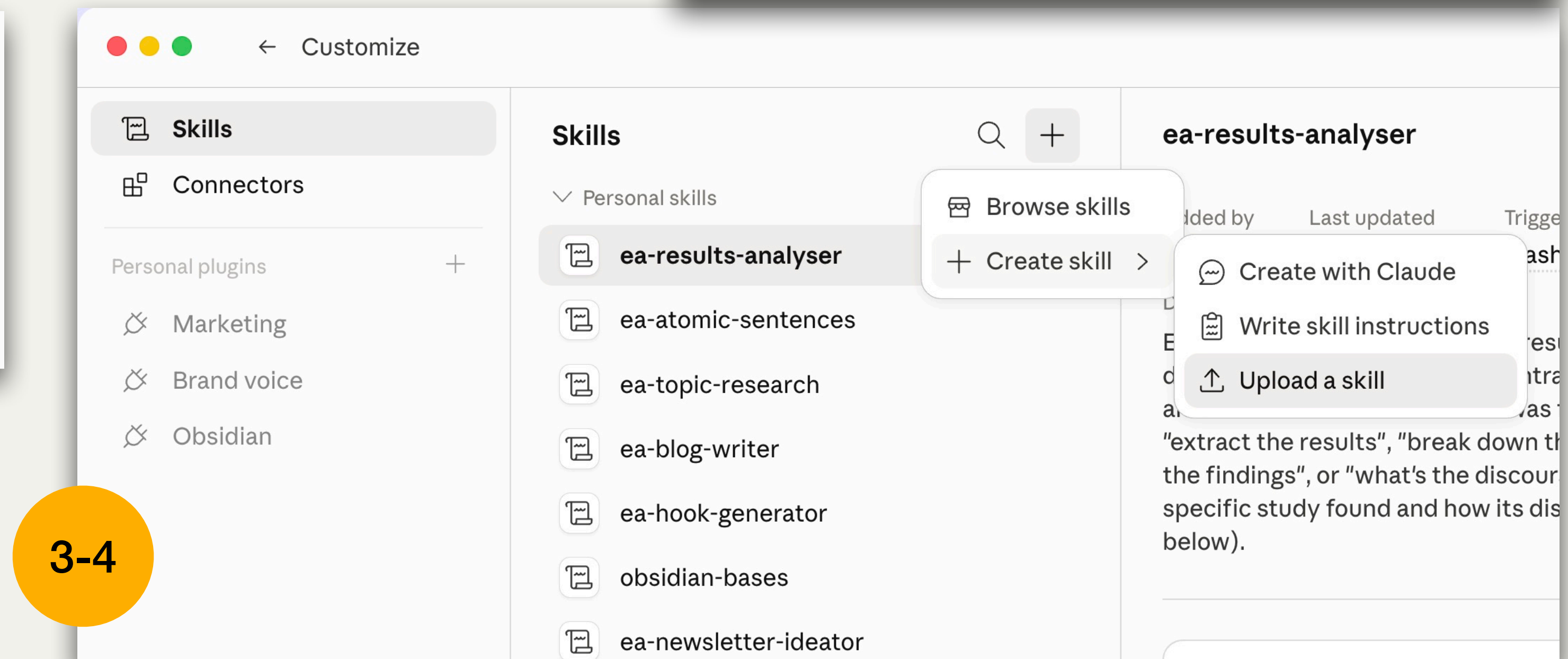
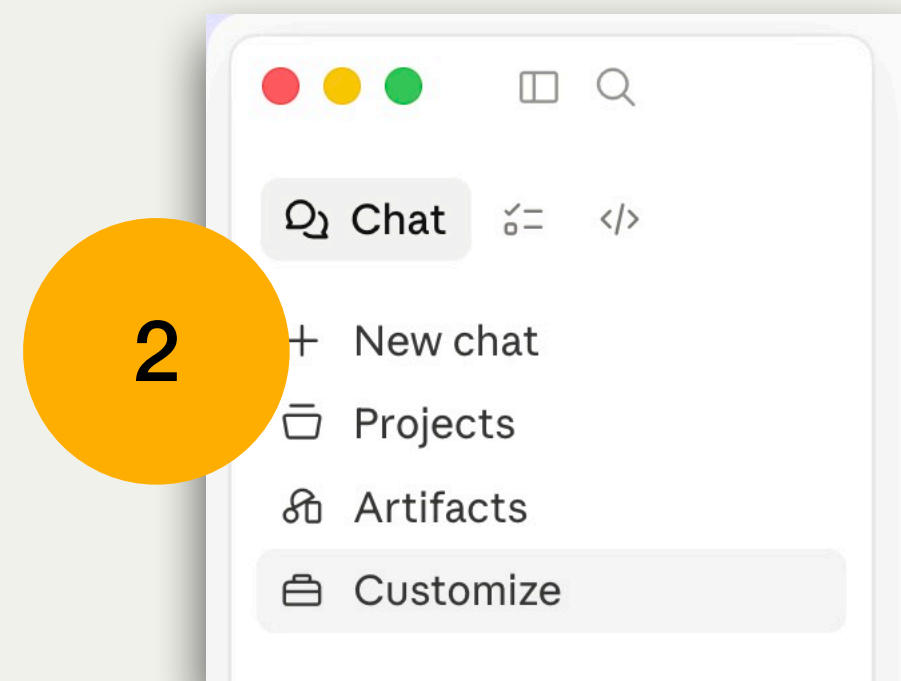
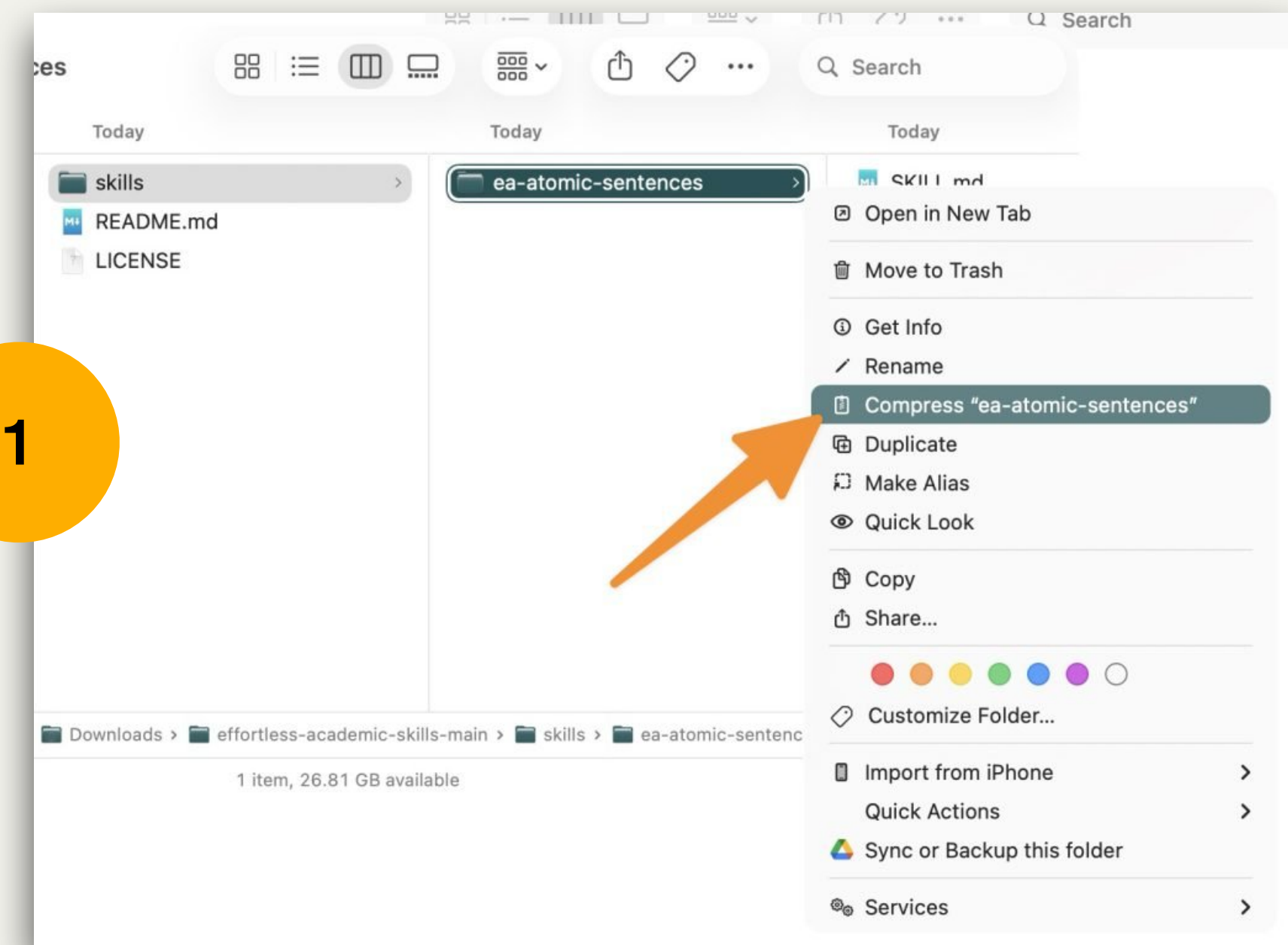
Navigate to Plugins > Skills and check if the skills
Show up in the personal section.



Installing Skills - Open Claude Desktop

1. Compress the skill folder into a zip file (one per skill)
2. In the Claude app select “Customize”
3. Click Skills
4. In the top right use the “+” button to upload a skill
5. Upload the zip file you created in 1.

Claude Desktop has different skills for the “chat” and “cowork” sections, you might need to reinstall the skill multiple times to use it everywhere.



To use a skill in Claude or Codex simply type in a / followed by the name of the skill.

A pop-up will appear to let you select the skill.

 **ea-blog-writer**

 **ea-social-media**

 **ea-hook-generator**

 **ea-topic-research**

 **ea-atomic-sentences**

 **ea-results-analyser**

 **ea-lecture-structurer**

 **ea-newsletter-ideator**

/ea|

+

Sonnet 4.6 ▾



Extracts atomic sentences from academic papers — identify every cited paper, compress each citation into a single logical claim, group them by sub-topic, and summarize each group. Use this skill whenever the user pastes academic text or uploads a PDF and wants to extract what cited papers actually claimed, map the literature, build a citation map, extract key arguments from a paper, atomize citations, or asks things like "what does each paper say?", "extract the key claims", "give me an atomic breakdown of this paper", "pull out all the references and their arguments", "summarize the citations by topic", or "elaborate on [Author Year]". Also trigger when the user is preparing a literature...

 Reasoning Medium

Skills

 **Ea Atomic Sentences** Extracts atomic sentences from academic papers — identify every cited paper... Personal

 **Ea Results Analyser** Extracts and analyses empirical results from primary research papers, summari... Personal

 **Plugin Creator** Scaffold plugins and marketplace entries System

 **Skill Creator** Create or update a skill System

 **Spreadsheets** Create and edit spreadsheet or excel files Personal

/ea

+

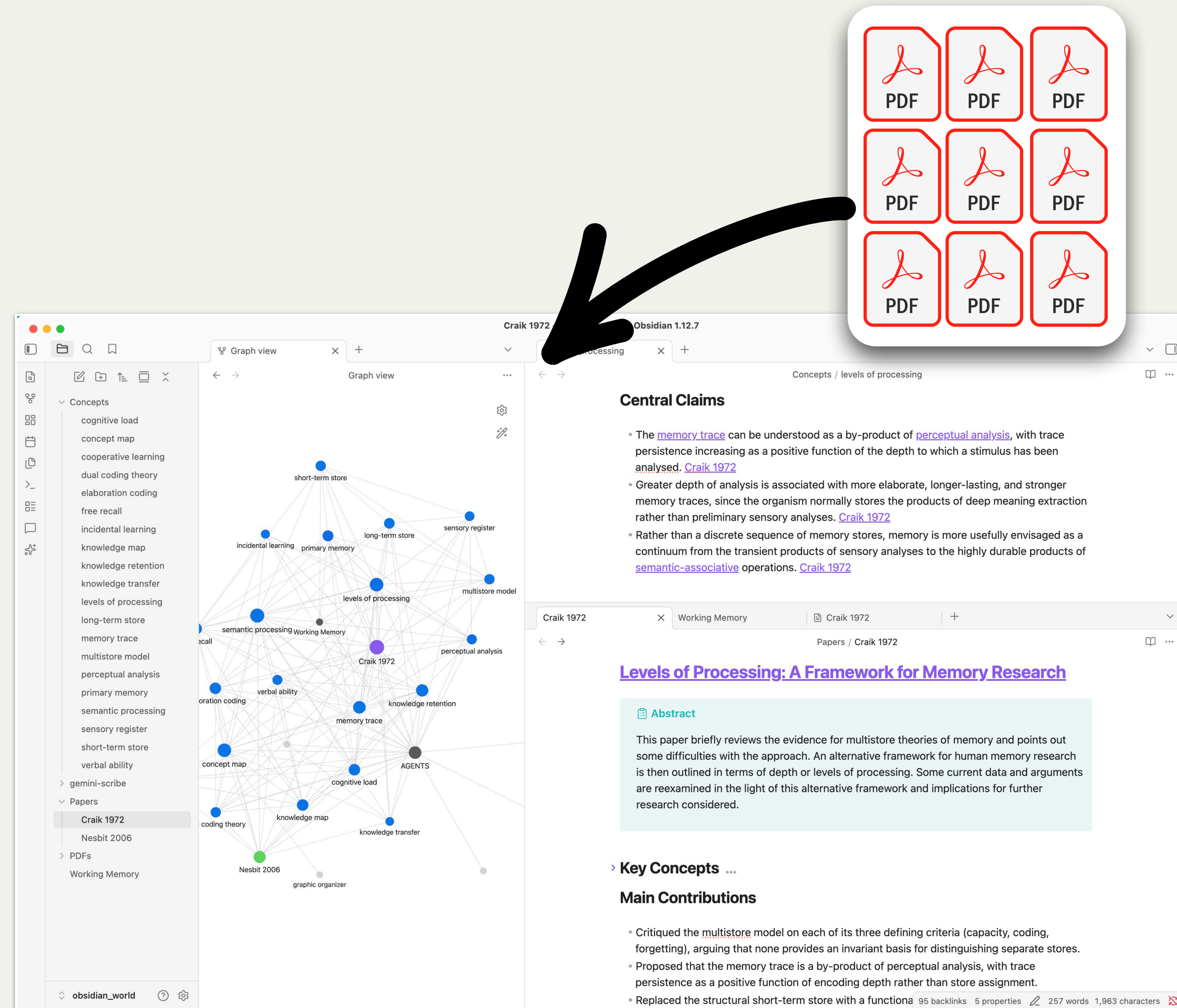
5.4 Medium ▾



Part 2: 5 Recommended Skills for Academics

Tip: Check out the [detailed blog post](#) on how to use this skill.



Link: <https://github.com/lnilya/effortless-academic-skills>

Purpose: Critically summarise only the results of an academic paper and identify parts of the discussion that contradict or support this result.

Why use it: AI Paper summaries are not useful, as you have the abstract. But the abstract usually does not report detailed results or what this means for the field. This skill solves that by telling you what a paper found and why it is relevant.

Usage: Type in /ea-results-summariser and drop in a paper (which has results, i.e. not a review paper).

Result 1

Leaf area in eucalypts is positively associated with mean annual precipitation (slope = 0.38, R^2 = 0.072) and weakly positively associated with mean annual temperature (slope = 0.0027, R^2 = 0.0011) across the datasets at higher

Importance: This climate relationships specimens could from conventional

Discussion

Wright et al. 2017 area associations providing the ben

An et al. 2021, su diverse plant taxa data.

Leigh et al. 2017, more readily due increases with ten

Result 2

The ML workflow achieved an overall precision of 82% (leaf masking model: 77%, improved to 82% with classifier), but a combined recall of only 34% and an F1 score of 48%, with training saturation reached at the volume of annotated images used.

Importance: This matters because the study prioritised precision over recall by design, aiming to minimise false positives rather than maximise capture of all leaves, as incomplete leaves would introduce greater downstream bias in the trait-climate analyses than missed leaves.

Discussion

Corney et al. 2012, exemplifies the result Suggested filtering the bottom 50% of leaves per species as a way to reduce bias from immature leaves, an approach the present study validated against and used to confirm slope convergence with traditional datasets.

Goëau et al. 2020, supports the result Recognised stochastic variation as an inherent challenge of working with herbarium sheets, supporting the authors' rationale that large dataset size provides a buffer against uncontrollable measurement error.

Willis et al. 2017, supports the result Identified spatial biases, sheet size limitations, and innate measurement errors as known issues in herbarium-based studies, contextualising the precision-first design strategy adopted here.

Link: <https://github.com/lnilya/effortless-academic-skills>

Purpose: Given a few bullet points with references (=the narrative), this skill creates three drafts of academic writing. Alternatively, you can give your written paragraph and ask for improvement.

Why use it: You know what you want to say in your paper, but formulating it in academic English is challenging. This skill fixes that by transforming ideas into academic writing. AI writes, but you control the narrative.

Usage: To create writing, provide a bullet point outline with a few references and ideas. This ensures that you have full control over the content. Then ask the AI to create a paragraph. The results will yield a very solid first draft without losing any control. Similarly, ask AI to criticise your writing to learn what can be improved. (You can use Skill #3 to extract atomic sentences)

[Data and algorithm problems of SDMs]

- SDMs and related models are prone to a series of methodological and data problems that the community is actively addressing (Soley-Guardia et al. 2024). Chief among them is the data quality
- Presence only is predominantly used, bc it is easier accessible, especially if data comes from herbarium or citizen science sources (e.g. iNaturalist) which lack absence records (Elith et al. 2020)
 - PO data has often strong sampling biases, accuracy problems (Graham et al. 2008) and can be spatially auto-correlated leading to degraded performance and misleading significance of explanatory variables (Dormann 2007; Segurado et al. 2006)
 - Typically data is thinned (by removing similar nearby observations), which decreases biases and auto-correlation issues, but decreases the available data and in some cases can worsen performance (Gaul et al. 2020; Ten Caten and Dallas 2023).
 - A better choice is presence-absence data, or a combination of the two (Simmonds et al. 2020). This typically involves forest inventory datasets, which however can also be prone to its own errors:
 - But also problems with in forest inventory datasets stemming from spp misidentification (around 5.9% (Scott & Hallam, 2003)), species detection (observer error) or plot relocation errors when remeasuring permanent plots (~15% and 21% resp. (Verheyen et al., 2018))
 - They cant be eliminated from past data and are often ignored, even thourh eliminating outliers is possible with a dedicated algorithm which can improve performance (Liu et al. 2018; Halladin-Dabrowska et al. 2019) but is rarely used.

Paragraph 1 — Data quality: presence-only records

Function: Introduces the dominant data problem in SDMs — spatial autocorrelation — and the costs of standard remedies

Speculative variant

Species distribution models are systematically undermined by the structure of available occurrence data. Presence-only records predominate because herbarium

Safe variant

Data quality in species distribution models is closely tied to the structure of available occurrence data.

Assertive variant

The dominance of presence-only data in species distribution modelling is a structural limitation, not a neutral methodological choice. Because herbarium collections and citizen science platforms such as iNaturalist generate occurrence records without absences, most SDMs are built on data that is biased, spatially autocorrelated, and locationally imprecise (Elith et al. 2020; Graham et al. 2008). Spatial autocorrelation in particular degrades model performance and inflates the apparent significance of predictor variables (Dormann 2007; Segurado et al. 2006). Spatial thinning partially addresses these problems but at the cost of data volume,

Link: <https://github.com/lnilya/effortless-academic-skills>

Purpose: Extract factual statements from papers (primarily from the intro and discussion) and break them into a series of logical statements with a reference, which can be reused for your paper.

Why use it: The easiest way to write academic papers is to think of each paper as a combination of factual “atomic” statements. Multiple statements fit together like LEGO pieces into a narrative. This skill lifts these LEGO pieces out of existing papers for you to recombine in your own.

Usage: Type in /ea-results-summariser and drop in a paper (which has results, i.e. not a review paper).

Make sure to double-check the output as AI can make mistakes.
More Info on the atomic-sentence method.

Scale-dependence and heterogeneity in biodiversity change

McGill et al. 2015

Biodiversity is complex, multidimensional, and scale-dependent.

Chase et al. 2018

Biodiversity change varies with the spatial scale of observation.

Sax and Gaines 2003

Local richness can remain stable even as regional species composition changes.

Thomas 2013

Observed biodiversity trends vary by scale; global declines coexist with local stability.

Primack et al. 2018

Local-scale biodiversity responses are variable and context-dependent.

Hillebrand et al. 2018

Local species richness (α -diversity) is a common metric but may obscure underlying change.

Gonzalez et al. 2016

There is wide spatial variation in biodiversity change estimates.

Cardinale et al. 2018

Local biodiversity responses to human activity are context-dependent.

Vellend et al. 2013

Local richness shows little net directional change over time.

Dornelas et al. 2014, 2019

No consistent global trend in local species richness; changes are compositionally

Blowes et al. 2019

Species turnover often dominates over net richness loss at local scales.

Conclusion: Biodiversity change is highly heterogeneous across space and time, with local-scale metrics often failing to capture broader patterns.

ECOG

Research article

The undetectability of global biodiversity trends in local species richness

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²Inst. of Biology, Martin Luther Univ. Halle-Wittenberg, Halle (Saale), Germany
³Dept of Life Sciences, Natural History Museum, London, UK
⁴Dept of Life Sciences, Imperial College London, Ascot, UK
⁵UN Environment Programme World Conservation Monitoring Centre, Cambridge, UK
⁶CIBIO (Research Centre in Biodiversity and Genetic Resources) – In. Porto, Vairão, Portugal

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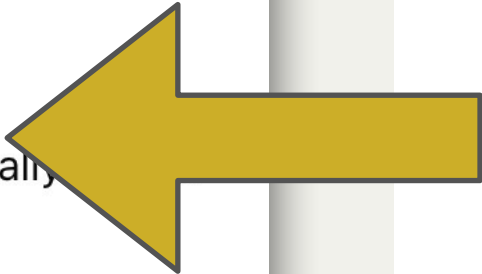
Although species are distributed unevenly across space, detecting trends in local species richness is difficult. Here, we use a novel approach to detect trends in local species richness (error of the measurement) as a proxy for the taxonomic richness of biodiversity and trends. We found that local species richness is not a good proxy for detecting trends in biodiversity. We found that local species richness is not a good proxy for detecting trends in biodiversity. We found that local species richness is not a good proxy for detecting trends in biodiversity.

Keywords: bias, measurement error, species richness, taxonomic richness

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www.ecography.org

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Oikos
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Link: <https://github.com/kepano/obsidian-skills>

Purpose: Create Obsidian Bases, which is a technical but extremely powerful feature of Obsidian.

Why use it: Bases are a feature that makes the digital note-taking tool Obsidian the best choice for academics (free, local, flexible, mature). But most newcomers are challenged by creating a base. This skill solves that by scanning what's in your vault and creating bases you ask for.

Usage: Ideally, use within the context of a project in Codex or CoWork; tell AI to scan your notes and create a base that summarises them in the way you need.

Tip: Obsidian is a tool which stores knowledge (notes and PDFs), just like Wikipedia, but locally for your personal research. Click here to learn more about it.



Papers by Year - Wetlandproject - Obsidian 1.12.7

TIP Discussion Para 2 | XAIP Story | PFT_Literature_Canvas | Papers by Year

1. Primary Sources / Papers by Year

Papers by Year 97 results

File	Title	Journal	Rating	Year	Summary
Year 2025					
Bukosa 2025(✓)	Inverse modelling of New Zealand's carbon ...	Atmos. Chem. Phys.	1/5	2025	Irrelevant. Model Co2 fluxes ar
Cambron 2025(✓)	Plant nutrient acquisition under elevated CO...	Nat. Clim. Chang.	4/5	2025	Review of what plant strategie
Diaz 2025(✓)	Plant functional traits and the entangled phe...	Funct. Ecol.	4/5	2025	Supporting the idea that phen
García Criado 2025(✓)	Plant diversity dynamics over space and tim...	Nature	5/5	2025	Alpha and beta diversity did no
Lubiana Botelho 2025(✓)	Calibrated ecosystem models cannot predict...	Ecol. Lett.	4/5	2025	Ecosystem models are too vag
Miyaji 2025(✓)	Interpreting ML in ecology: A RAG-based ap...	Anais do XVI workshop de computação aplic...	3/5	2025	Use an LLM Agent and some p
Year 2024					
Guo 2024(✓)	Using machine learning to link climate, phylo...	J. Ecol.	3/5	2024	Use ML method to create a da
Iban 2024(✓)	SHAP-driven eXplainable Artificial Intelligenc...	Remote Sens. (Basel)	2/5	2024	Use Explainable AI to identify v
Johnson 2024(✓)	Revealing uncertainty in the status of biodiv...	Nature	5/5	2024	Uncertainty in biodiversity esti
Montràs-Janer 2024(✓)	Anthropogenic climate and land-use change...	Nat. Ecol. Evol.	5/5	2024	Show that species richness inc
Moss 2024(✓)	Drought as an emergent driver of ecological ...	Bioscience	5/5	2024	Review of drought effects on v
Ricotta 2024(✓)	What do functional diversity, redundancy, rar...		4/5	2024	Mathematical exploration of fu
Soley-Guardia 2024(✓)	Top ten hazards to avoid when modeling spe...	Ecography (Cop.)	4/5	2024	Collection of typical mistakes p
Song 2024(✓)	Explainable artificial intelligence to interpret ...	I-guide forum 2024: Convergence science a...	3/5	2024	Train an SDM and explain cont
van der Sande 2024(✓)	Tropical forest succession increases tree tax...	Glob. Ecol. Biogeogr.	4/5	2024	FRic increases, FEve decrease
van Steenderen 2024(✓)	Climate covariate selection influences MaxE...	Ecol. Modell.	3/5	2024	Took 19 Bioclim variables and
Velazco 2024(✓)	How far can I extrapolate my species distrib...	Ecography (Cop.)	5/5	2024	Present the Shape method

Link: <https://github.com/paramchoudhary/resumeskills>

Purpose: Polish your academic CV for grant applications, PhDs or other occasions.

Why use it: A strong, well-formatted CV can make the difference between a new job and a failed attempt.

Usage: Upload your existing CV to receive feedback. The output will be in Markdown, and you'll need to convert it back to MS Word.

Output Format

When creating an academic CV:

```
# ACADEMIC CV STRUCTURE FOR [NAME]

## Recommended Section Order
[Based on position type and field]

1. [Section]
2. [Section]
...

## Section Content

### Education
[Formatted education section]

### Publications
[Formatted with appropriate style for field]

### [Other Sections]
[Formatted content]

---

## Formatting Notes
- [Field-specific conventions to follow]
- [Style guide recommendations]

## Things to Add/Update
- [ ] [Missing item]
- [ ] [Item needing update]
```

Link: <https://github.com/lnilya/effortless-academic-skills>

Purpose: Given papers that ideally you have written, will create an introduction covering all topics mentioned in these papers.

Why use it: When writing a dissertation by publication, simply upload your papers, and the introduction outline is ready. Saves you from missing any important topics.

Usage: Upload papers and mention the skill. The first step will ask you to select topic areas, and the second step will suggest an outline for the introduction.

Tip: If working on a lit-review paper, you could also upload papers from others and extract all the concepts they mention. You will need a solid grasp on the subject-matter but the skill can help you break down your review paper into topics.

Plan academic introduction ...

Ati.docxBDP.docxTIP.docxEa Intro Planner

16 previous messages >

I extracted the three DOCX files, built the Phase 1 topic hierarchy, and saved it here: [intro_workspace.json](#).

Topic Map

- 1. Climate-driven biodiversity redistribution**

Key refs: Lawlor 2024; Parmesan 2006; Lenoir & Svenning 2015

 - Upward and poleward shifts
 - Novel ecological conditions
 - Context-dependent redistribution
 - Response lags
- 2. Drivers beyond temperature**

Key refs: Taheri 2021; Crimmins 2011; Louthan 2015

I updated [intro_workspace.json](#) and built the intro around **range shifts + biome-specific responses**, with methods kept mostly as positioning rather than foreground.

P1 — Domain And Stakes

Climate change is reorganising forest biodiversity in ways that threaten ecosystem function before losses are visible.

- Establish forest redistribution, because composition is shifting under novel climates.
- Link reorganisation to ecosystem function, because forests regulate carbon and resilience.
- Note threshold risks, because gradual warming can produce abrupt community change.
- Frame prediction as urgent, because conservation decisions need early warning signals.

→ *The problem is not simply whether forests change, but how early that change can be detected.*

Refs: Lawlor (2024); Kerr (2025); Trumbore (2015); Carnicer (2019); Trisos (2020) | Search: "climate change forest community reorganization ecosystem function"

Ilya Shabanov, The Effortless Academic

Not your average scientist: I spent 12 years in industry, co-founded a brain training startup (NeuroNation), ran a web design firm and only at 36 decided to do a PhD in biology. At this point, I didn't know much about biology – yet 6 months later, I had a paper in peer review and a glowing recommendation from my supervisor:

“Ilya’s ability to assimilate the knowledge he had read and gleaned from our conversations, [...] was in my view remarkable [...] Frankly, I feel like Ilya can already intellectually “spar” with me at the same level as my most senior graduate student of 4 years.”

My secret to learning very fast is not talent. It's “Knowledge Management”, something I did routinely as a CEO, just like when we play chess, the better strategy always wins.

I founded The Effortless Academic in 2022 to share my insights on digital note-taking. As the AI explosion unfolded, I grew the site into one of the largest resources for academic use of AI.



I am currently working on finishing a PhD in Ecology in 2026.

How Can Effortless Academic Help You?

The Effortless Academic offers various free and paid courses for academics to:

- Learn about new online tools for research
- Understand and use AI efficiently and ethically
- Store ideas and information digitally
- Organise large academic projects & meetings

If you want to get started, sign up for the two **free courses** on

» [Knowledge Management and Digital Notes](#)

» [Literature Reviews in the age of AI](#)

If you want more info on the in-depth courses, visit the website:

EffortlessAcademic.com





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